

Airbyte: Airbyte provides open source ELT connectors for databases, SaaS apps, and APIs. It offers 300+ ready-to-use connectors and a simple YAML-based configuration model. Its modular design makes it easy to extend and integrate into modern data platforms.

Singer (Taps & Targets): This tool defines a simple, open standard for data extraction and loading. Taps pull data from sources, and Targets load it into destinations. It's lightweight, scriptable, and ideal for custom or niche integrations.

Apache Kafka: Kafka is a distributed event-streaming platform used for real-time data ingestion and messaging. It provides durable logs, high throughput, and replayable streams. It's foundational for event-driven architectures and ML feedback loops.

Apache Airflow: This workflow management tool is used to schedule complex data pipelines from ingestion to training. Airflow orchestrates complex workflows using Python-based DAGs. It handles scheduling, retries, dependencies, and lineage and is widely used for ETL, ML pipelines, and cross-system automation.

Apache Spark: This distributed compute engine for large-scale data processing supports SQL, machine learning, streaming, and graph analytics. It's in-memory execution makes it significantly faster than traditional MapReduce.

dbt Core: This tool enables SQL-based data transformations with software-engineering best practices. It brings versioning, testing, documentation, and lineage to analytics pipelines. It's ideal for building curated datasets and ML-ready tables.

Great Expectations: Great Expectations is a data quality and validation framework. It lets teams define 'expectations' for data and automatically test them. It generates human-readable reports and integrates seamlessly with pipelines.

JupyterLab: This is an interactive development environment for notebooks, code, and data exploration. It supports multiple kernels (Python, R, Julia) and rich visualisations. It's the default workspace for data scientists.

Apache Superset: Superset is an open source BI and dashboarding platform. It offers SQL exploration, charting, and interactive dashboards. It's lightweight, cloud-friendly, and ideal for analytics teams.

Jenkins: This widely used open source CI/CD automation server supports pipelines-as-code, plugins, and integrations with any tool. It's often used to automate model packaging, testing, and deployment.

Docker: Docker packages ML code, dependencies, and models into portable containers. It ensures reproducibility across dev, test, and production. It's essential for Kubernetes-based ML platforms.

Kubeflow Pipelines: This orchestrates end-to-end ML workflows on Kubernetes. It provides reusable components, metadata tracking, and experiment management, and is ideal for scalable, cloud-native ML training.

PyTorch/TensorFlow/scikit-learn: These are the core open source ML frameworks.

- **scikit-learn:** Classical ML, fast prototyping
- **PyTorch:** Flexible deep learning, research-friendly
- **TensorFlow:** Production-grade deep learning with strong ecosystem support

MLflow Model Registry: Model Registry stores, versions, and manages ML models. It supports lifecycle stages (staging → production), lineage, and approvals. It integrates with MLflow tracking and serving tools.

Seldon Core: Seldon Core deploys and serves ML models on Kubernetes. It supports REST/gRPC endpoints, canary releases, A/B tests, and explainers. It's highly scalable and production ready.

Prometheus + Grafana: Used for monitoring, these visualise performance metrics and identify anomalies. Prometheus collects time-series metrics, and Grafana visualises them. Together, they provide observability for ML services (latency, throughput, errors).

ELK Stack (Elasticsearch, Logstash, Kibana): The ELK stack handles centralised logging and search. Logstash ingests logs, Elasticsearch indexes them, and Kibana visualises them. ELK is widely used for debugging ML pipelines and services.

Kafka (Feedback Loop): Kafka captures predictions, outcomes, and user interactions for retraining. It enables event-driven ML and continuous learning.

LLMOps: The future of MLOps

Generative AI models have become mainstream in enterprise development. Organisations are now shifting toward LLMOps, or large language model operations, to manage autonomous agentic workflows and enforce safety and governance. LLMOps spans prompt management, vector-store orchestration, fine-tuning pipelines, predictive modelling, integrated safety layers, and continuous evaluation, enabling reliable, governed, and cost-efficient scaling of genAI systems. This evolution requires traditional MLOps teams to expand their skills and operate as LLMOps practitioners to ensure generative outputs remain secure, aligned, and trustworthy.

The core capabilities of LLMOps are:

Prompt management and versioning: Track, test, and optimise prompts as first-class artifacts.

RAG (retrieval-augmented generation) operations: Manage vector databases, embeddings, chunking strategies, and retrieval pipelines.

Model lifecycle management: Handle fine-tuning, model selection, model switching, and rollback.